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|----------|------------|--|----------|-----|
| 3 | (a) | (i) the total momentum of a system (of interacting bodies) remains constant
provided there are no resultant external forces / isolated system | M1
A1 | [2] |
| | | (ii) elastic: total kinetic energy is conserved, inelastic: loss of kinetic energy
[allow elastic: relative speed of approach equals relative speed of separation] | B1 | [1] |

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- (b) (i) initial mom: $4.2 \times 3.6 - 1.2 \times 1.5$ (= 15.12 – 1.8 = 13.3) C1
 final mom: $4.2 \times v + 1.5 \times 3$ C1
 $v = (13.3 - 4.5) / 4.2 = 2.1 \text{ m s}^{-1}$ A1 [3]
- (ii) initial kinetic energy = $\frac{1}{2} m_A(v_A)^2 + \frac{1}{2} m_B(v_B)^2$
 = 27.21 + 1.08 = 28(.28) M1
 final kinetic energy = 9.26 + 6.75 = 16 M1
 initial KE is not the same as final KE hence inelastic A1 [3]
provided final KE less than initial KE
 [allow in terms of relative speeds of approach and separation]